A

Exclusively recessive (OMIM, HPO)
2,470 genes with an organ profile

Skeletal dysplasia nosology 2023
736 entries, 542 genes

Organ profile per gene
fraction of HPO terms per organ, ≥ 0.2

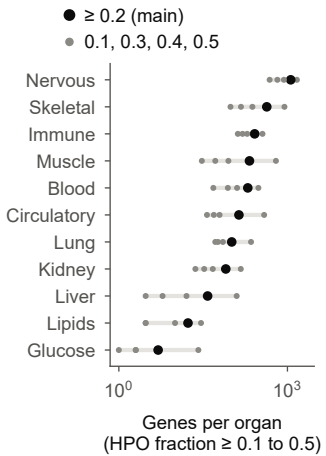
Dysplasia genes in Genebass
282 AR, 166 AD, 46 AD/AR, 36 other

Control gene sets
recessive non-skeletal (n = 1,076); no Mendelian disease (n = 14,596)

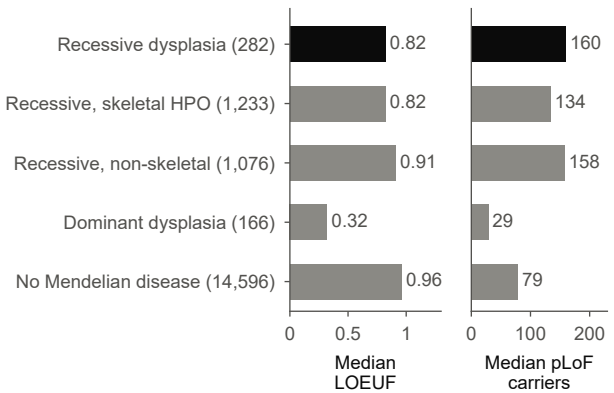
B

Organ (genes at ≥ 0.2)	UKB traits (31; primary first)
Blood (197)	Hemoglobin, MCV, RBC, Platelets
Circulatory (137)	Systolic BP
Glucose (5)	HbA1c, Glucose
Immune (261)	WBC, Lymphocytes
Kidney (80)	Cystatin C, Creatinine, Urate, Urea, Phosphate
Lipids (17)	LDL, HDL, Triglycerides, Cholesterol
Liver (38)	ALT, AST, GGT, Bilirubin, Albumin
Lung (102)	FEV1, FVC
Muscle (211)	Grip strength
Nervous (1,144)	Fluid intelligence, Reaction time
Skeletal (429)	Height, Sitting height, Heel BMD

C



D



E

